

Topic

- Association Rule Learning
 - Market Basket Analysis
 - Support
 - Confidence
 - Lift
 - Apriori Algorithm
 - FP-Growth (concept only)
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Class Objective

Students will:

- Understand how products/items are associated
 - Learn Market Basket Analysis
 - Understand:
 - Support
 - Confidence
 - Lift
 - Learn Apriori concept practically
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1. CLASS START (Hook)

Start like this:

“Have you noticed...

Amazon says: ‘People who bought this also bought this?’”

 Pause

Then say:

“That is Association Rule Learning.”

2. WHAT IS ASSOCIATION RULE LEARNING?

Definition

Association Rule Learning finds relationships between items in data.

Simple Meaning

“It discovers which items occur together frequently.”

3. 🛒 MARKET BASKET ANALYSIS

Example

Customer buys:

- Bread
- Butter

👉 Many customers buy both together

So system learns:

Bread → Butter

💡 Teaching Line

“Association learning finds hidden buying patterns.”

4. 📁 REAL-LIFE APPLICATIONS

- ✓ Product recommendation
 - ✓ Shopping websites
 - ✓ Netflix recommendations
 - ✓ Medical diagnosis patterns
 - ✓ Website click analysis
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5. 📖 IMPORTANT TERMS

◆ SUPPORT

Meaning

How frequently items appear together.

Formula Concept

Support = Transactions containing item / Total transactions

Example

Bread appears in 4 out of 10 transactions

Support = $4/10 = 0.4$

Teaching Line

“Support measures popularity.”

◆ CONFIDENCE

Meaning

Probability that item B is purchased when item A is purchased.

Example

If customer buys bread,
how often they buy butter?

Teaching Line

“Confidence measures reliability of rule.”

◆ LIFT

Meaning

Measures strength of association.

Interpretation

Lift Meaning

>1 Positive association

=1 No relation

<1 Negative relation

Teaching Line

“Lift tells whether association is truly meaningful.”

Definition

Apriori finds frequent item combinations.

Simple Meaning

“Find common item groups first, then generate rules.”

7. HOW APRIORI WORKS

Steps

1. Find frequent items
 2. Find frequent item combinations
 3. Generate rules
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8. LIMITATION OF APRIORI

-  Slow for large data
 -  Many combinations
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 Solution:

FP-Growth

9. FP-GROWTH (Concept Only)

Definition

More efficient association rule algorithm.

Difference

Apriori	FP-Growth
Slower	Faster
Many scans	Compact tree structure

 No deep coding needed

10. 🎯 INTERVIEW CONCEPTS

Q1: What is Association Rule Learning?

👉 Finding relationships between items.

Q2: What is Support?

👉 Frequency of item occurrence.

Q3: What is Confidence?

👉 Reliability of association rule.

Q4: What is Lift?

👉 Strength of association.

Q5: Difference between Apriori & FP-Growth?

👉 FP-Growth is faster and more efficient.

11. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Find item associations using Apriori

OUTPUT INTERPRETATION

Explain:

- Frequent itemsets → common combinations
 - Rules show:
 - antecedents
 - consequents
 - support
 - confidence
 - lift
-

👉 Example:

Bread → Butter

Means:

Customers buying bread also buy butter.

Association Rule Learning helps businesses discover hidden product relationships and improve recommendations.”